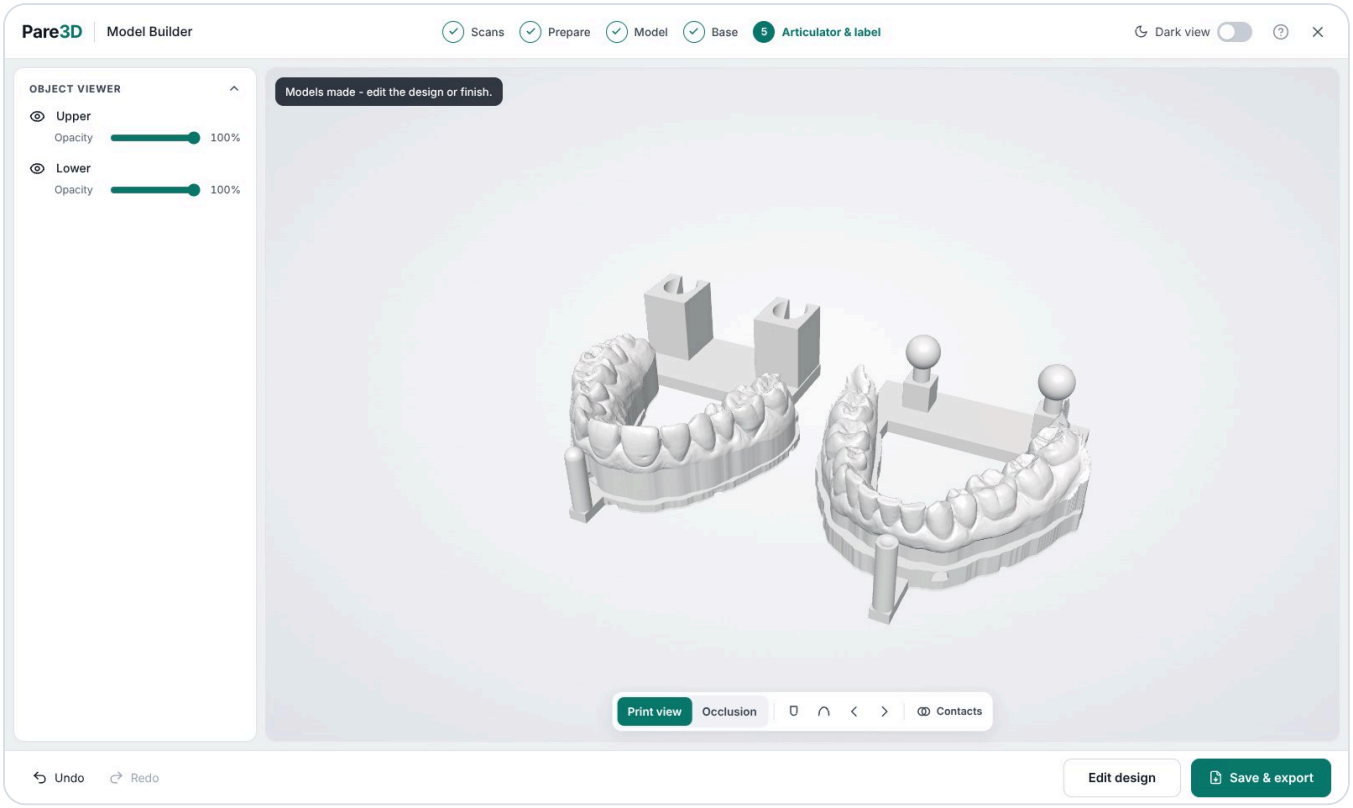
Pare 3D · Dental Studio · November 2026

The Model Builder Guide

From intraoral scans to printed study, orthodontic, aligner, crown-and-bridge and articulated models, in five steps.



An upper and lower pair made from three scan files, with the printed articulator, laid out ready to print.

Screens are from the current Pare 3D build at 1600 × 900. The scans used are a full upper arch, a full lower arch and a buccal bite scan from the author's own mouth.

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The short way

Dental Studio › Model Builder › **New case** › drag the upper, lower and bite scan files onto the window › **Next** › trim each arch and press **Accept** › choose what the model is for › choose the base thickness › **Separate models** or **Articulated models**, type a label › **Make models** › **Save & export**. The rest of this guide explains each choice.

1 What the Model Builder does

The Model Builder turns raw intraoral scans, which are open surfaces that stop where the scanner stopped, into closed models ready to print. Each model gets a base, and a pair can be set in the bite with a printed articulator.

It does the work of a lab technician or a model-builder program, in five steps shown across the top of the screen. Each step asks for one decision and fills in sensible settings for the rest. You can go back to any step at any time.

What is never changed

- **The teeth.** The scanned tooth surfaces are carried into the model exactly as the scanner recorded them. Bases, pins and the articulator are built around the scan, never into the teeth.
- **The bite.** The jaw relationship your scanning software exported is kept as it is. Nothing is moved unless you trim the scan yourself.
- **Your scan files.** The originals are never overwritten. The case keeps its own copy of the scans with every setting, so the design can be reopened and changed later.

What you get

- An upper and a lower model, or a single arch, as closed solids with a flat base, laid out to print.
- For a pair, optionally, a printed articulator: ball condyles behind the last molars that drop into cups under the upper, and an optional pin to hold or open the bite.
- A label engraved on the base, which also names the saved files.
- For hollow and honeycomb models, drain slots at the base so the model prints safely flat on the build platform (chapter 8).

2 Before you start: scans and files

File types

The Model Builder reads **STL**, **PLY** and **OBJ** files, which is what intraoral scanners export (Medit, 3Shape TRIOS, iTero, Primescan and others). Colour is not needed. The scans are shown and printed in grey.

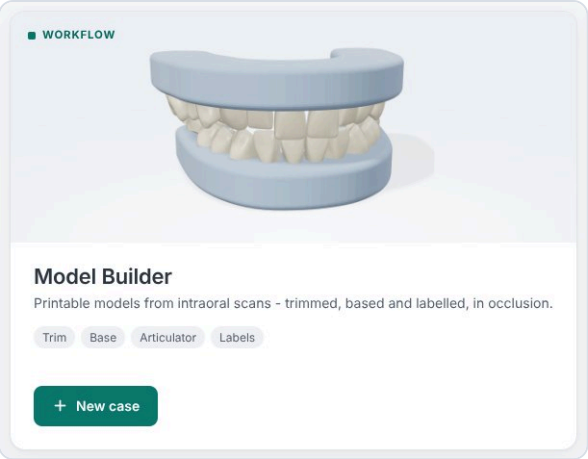
What to export from the scanner

- **The upper arch** and **the lower arch** as separate files, in the scanner's own coordinates. Do not move or re-align them in other software first. The bite depends on the coordinates the scanner gave them.
- **One or two bite scans** (buccal registration scans), if your scanner exports them. They are used to check the bite and can be shown over the models. Two bite scans, one each side, are the most reliable [13-15].
- A full arch or a quadrant. A quadrant against a full arch is handled as a pair.

Scan quality that matters

- **Enough depth below the teeth.** The base is built from where the scan ends. Scan a few millimetres of gum beyond the gingival margin all round.
- **No holes in the teeth.** A void in a tooth shows on the model. Rescan the area rather than relying on a fill.
- **A bite taken in maximum intercuspation**, with the patient biting firmly, on both sides where possible.

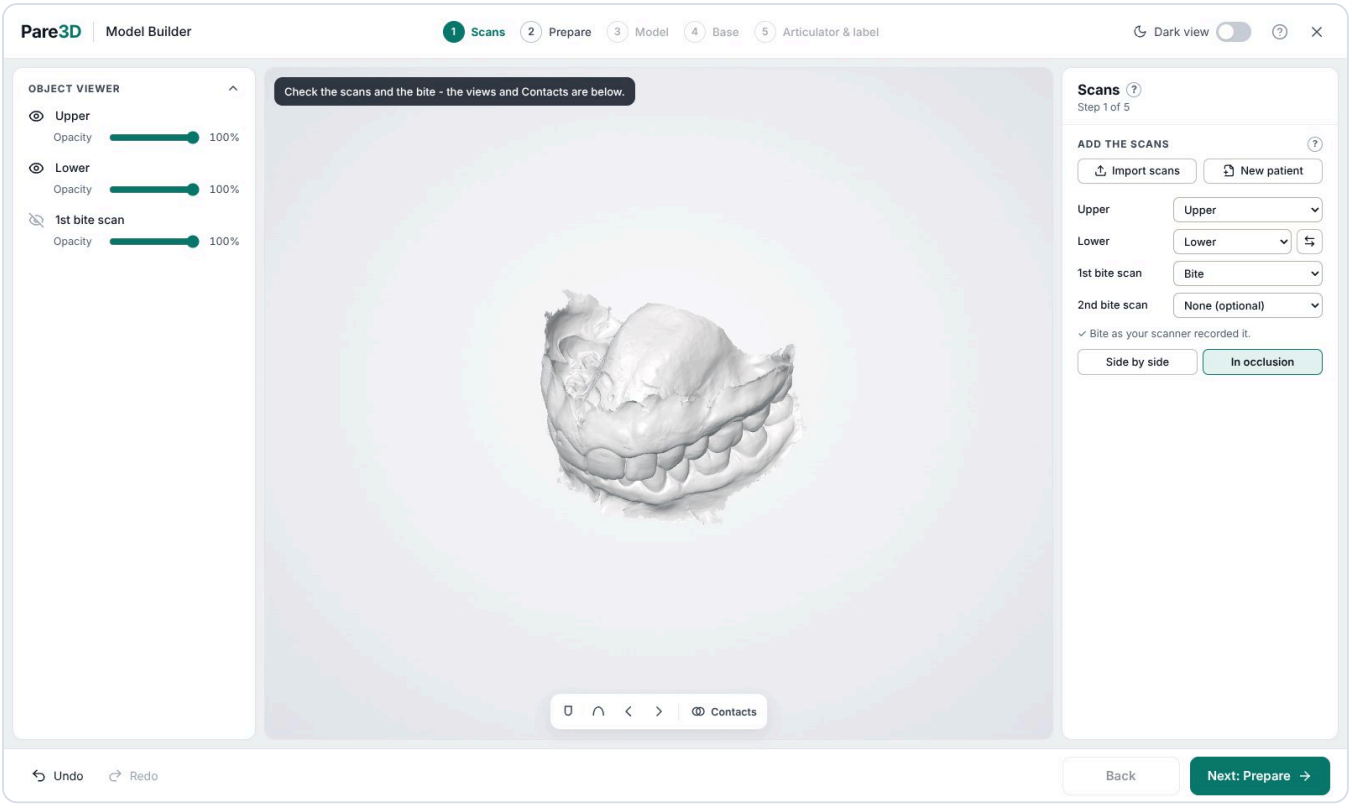
3 Starting a case, and the studio screen



The Model Builder card on the Dental Studio page.

1. On the home page, open **Dental Studio** at the top.
2. On the **Model Builder** card, click **New case**. The studio opens at step 1.
3. Your cases are listed under **Cases**. Each card shows the patient label, a **Model Builder** tag, and how far the case has got, for example *In progress: Prepare* or *Models made · Articulated*. Click a card to open the case where you left it.

The studio screen



The studio screen at step 1, with the scans loaded and shown in the bite.

Part of the screen	What it does
Steps (top centre)	Scans · Prepare · Model · Base · Articulator & label. The current step is filled. A finished step shows a tick and can be clicked to go back to it.
Dark view, **?** and **×** (top right)	Dark view puts a dark background behind the model, which helps with a pale scan. **?** opens help. **×** closes the studio and returns to the Dental Studio page. The case is kept.

Object viewer (left)	Each scan, and each bite scan, with an eye to show or hide it and an **Opacity** slider to see through it. Bite scans appear on step 1 only. Once the models are made, they are listed here too.
The view (centre)	The scans or models. Right- or middle-drag turns the view, the wheel zooms. A short hint at the top left says what to do at this step.
View bar (bottom of the view)	Front view, occlusal view, turn left and right, and **Contacts**, the bite contact map (chapter 5).
Panel (right)	The choices for the step you are on. Each heading has a **?** with a short explanation, and holding the pointer over a button shows what it does.
Undo, Redo, Back, Next (bottom)	Undo and Redo undo and redo the last change. **Back** and **Next** move between steps. Next is green when the step is complete. It never moves on by itself.

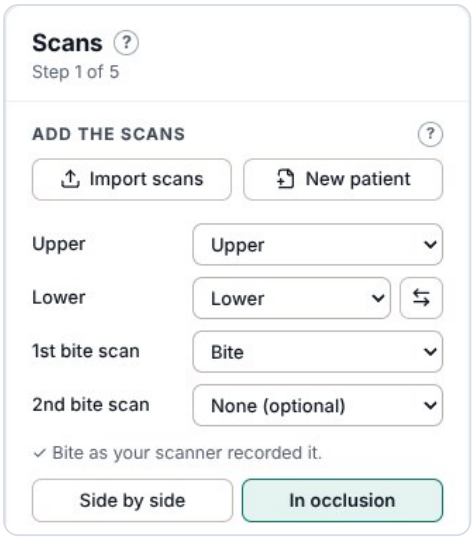
The right-click menu

Right-click anywhere in the view, on the model or beside it, for a menu that suits the step you are on. It holds the six standard views (front, occlusal, each side, lingual), **Bite contacts**, the viewing options (**Use the gimbal**, **Free rotate**, **Full rendering (slower)**) and, while you trim, the trim tools.

> **A large scan**
> If a scan has more triangles than the editor's limit for cutting, a notice offers to raise it. Choose **Not now** to carry on. The Model Builder works either way.

4 Step 1 - Scans

Bring in the upper and lower scans, and one or two bite scans if you have them.



The Scans panel once the files are in.

1. Drag the scan files onto the window, all at once or one by one, or click **Import scans** and choose them.
2. Pare 3D cleans each scan (loose fragments, specks and folded edges), turns it teeth-up and works out which is the upper, which the lower and which are bite scans. It fills in **Upper**, **Lower**, **1st bite scan** and **2nd bite scan** for you.
3. Check the four boxes. If the upper and lower are the wrong way round, press the swap button beside **Lower**, or choose from the lists.
4. Choose **Side by side** or **In occlusion** to look at the scans either way. This only changes the view.
5. Press **Next: Prepare**.

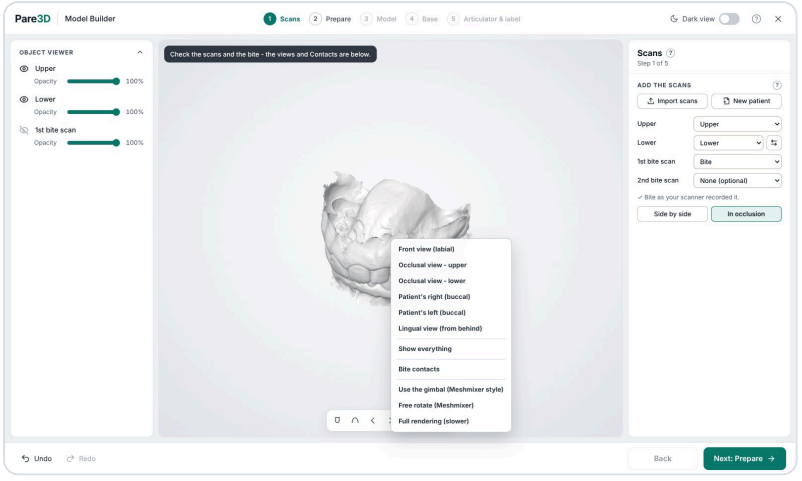
New patient clears the scans and starts again.

How the upper is told from the lower

The upper has the palate, or the bigger, flatter arch, and in the bite the upper incisors stand in front of the lower (the overjet). Where those disagree, the overjet decides. A quadrant is recognised by its length and is made as a quadrant.

Bite scans in the view

Bite scans are hidden at first, so they do not cover the arches. Click the eye beside **1st bite scan** in the Object viewer to show it. It is drawn in amber over the arches in occlusion, so you can see that it sits where the teeth are. Bite scans only appear at this step.



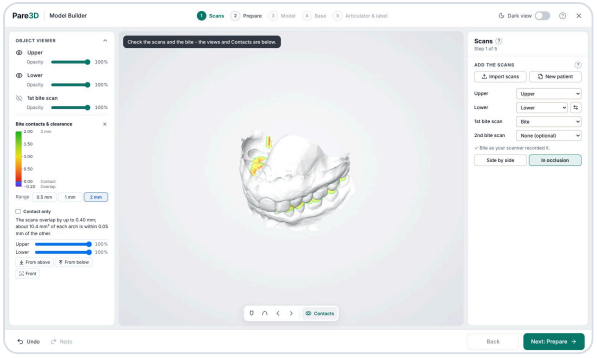
The right-click menu at step 1: the views, Bite contacts and the viewing options.

5 The bite, and checking the contacts

The bite is kept exactly as your scanner exported it. Step 1 tells you so: ✓ *Bite as your scanner recorded it.*

Intraoral scanners place the two arches in the bite when they process the buccal bite scans, and that placement is more accurate than a silicone or wax record [10]. Re-fitting the arches to the bite scans in another program adds error rather than removing it. The Model Builder therefore keeps the scanner's bite. Each bite scan is still checked against the arches. If one disagrees with them by more than 0.5 mm, an amber note says so, and you should check the occlusion from the front and both sides before making the models.

Bite contacts



Contacts on: the map on the teeth and its card at the left.

Press **Contacts** in the view bar, or choose **Bite contacts** in the right-click menu. Each arch is coloured by how far it is from the other:

- **Purple and blue**: the scans run into each other (overlap).
- **Red**: touching.
- **Green**: clear, up to the chosen range (0.5, 1 or 2 mm).

The card at the left gives the scale and a summary, for example how deep the scans overlap and how much of each arch is within 0.05 mm of the other. **Contact only** shows only the touching areas. The map shows a distance, not a pressure. A scan records no force.

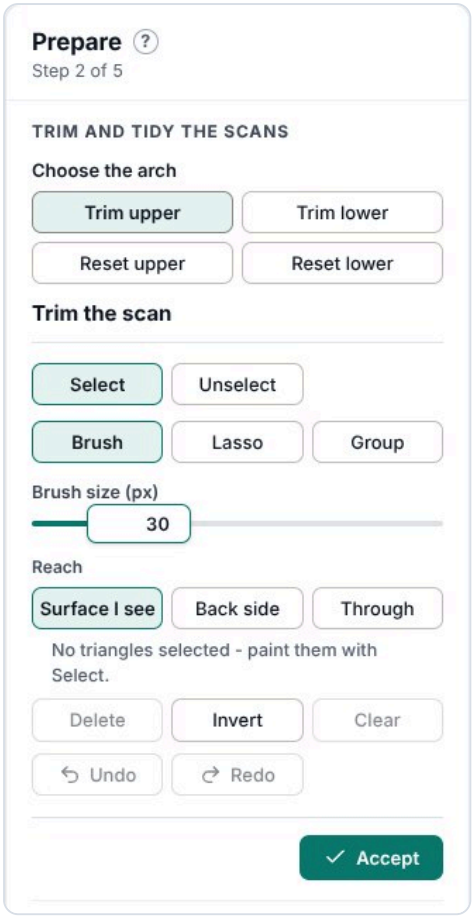
Contacts can be shown at every step, and on the finished models. They follow the **Print view** / **Occlusion** switch after making.

Overlap

A small overlap (a few hundredths of a millimetre) is normal. A deep or wide overlap means printed models would hold each other open there. Check the scans and rescan the bite if needed.

6 Step 2 - Prepare: trim the scans

Trim away scan tags, excess gum and anything you do not want on the model. Each arch is accepted before you move on.



Trimming the upper: the arch's button is green, and the trim tools sit underneath.

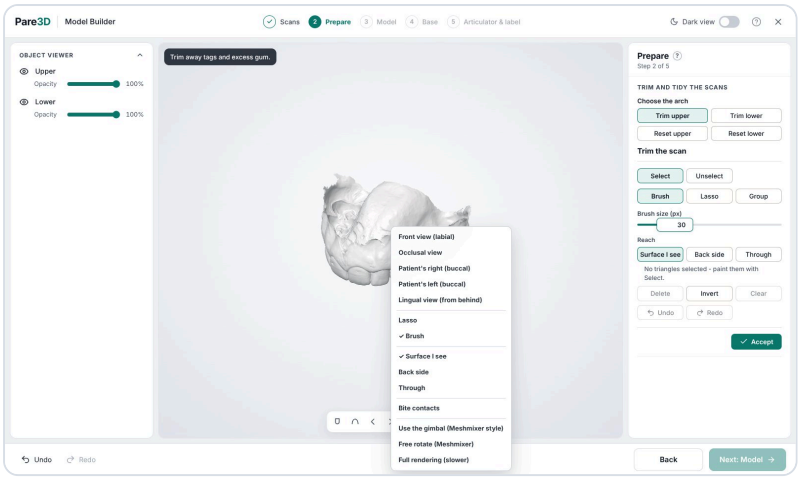
1. Press **Trim upper**. The button turns green and the trim tools open below it, in the panel.
2. Choose **Select** (or **Unselect** to take some back) and a shape:
   - **Brush**: paint over the surface. Set its size with **Brush size**.
   - **Lasso**: draw round an area.
   - **Group**: click a region to select all of it.
3. Choose the **Reach**:
   - **Surface I see** selects only the front surface, nothing hidden behind it.
   - **Back side** selects the side facing away from you, such as the underside of a scan.
   - **Through** selects every layer under the brush.
4. Press **Delete** to remove what is selected. **Invert** swaps the selection, and **Clear** lets go of it. **Undo** and **Redo** step through your trims.
5. Press **Accept**. The lower opens for trimming. Trim it the same way and press **Accept**.
6. When both arches are accepted, **Next: Model** turns green. Press it.

No trimming needed?
Open each arch and press **Accept** straight away. Accepting an arch as it is counts.

Lasso, Brush and the three reaches are also in the right-click menu while you trim, with **Delete selection**. **Reset upper** and **Reset lower** put an arch back as it was imported.

Contacts can be shown while you trim. The card appears at the left, so you can see whether a tag of gum you are removing touches the other arch.

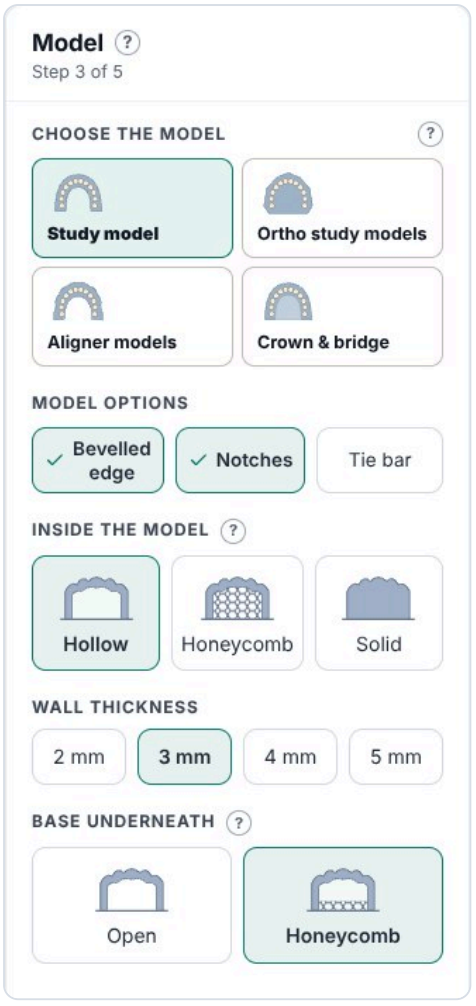
Never trim the teeth
Pare 3D will not cut or fill over the crowns of the teeth. If a selection reaches them, the trim stops and tells you how many triangles are on the crowns. Unselect them, or press again to go ahead. The teeth are what the model must reproduce.



While trimming, the right-click menu adds Lasso, Brush and the three reaches.

7 Step 3 - Model

Choose what the model is for. The settings that suit it are filled in. Every one can be changed.



The Model panel for a Study model.

Choose the model

Model	What it sets
Study model	A clean base a few mm round the teeth, hollow with a honeycomb in the base, 3 mm walls. Light and quick to print.
Ortho study models	The trimmed orthodontic base (square back, angled heels, pointed upper front, rounded lower front), solid, as a matching pair. No soft tissue is trimmed: records must not be altered [19].
Aligner models	Solid, about 18 mm tall overall, for thermoforming [1].
Crown & bridge	Solid, with a plate across the arch. The most accurate [2].

Model options

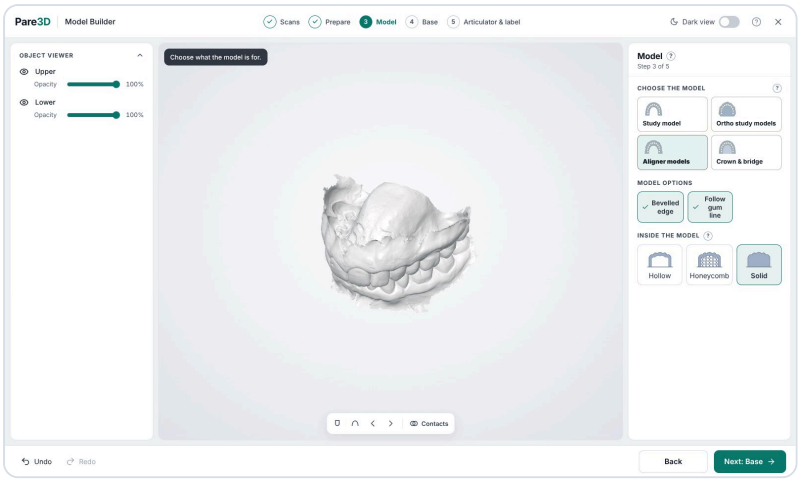
Option	What it does
Bevelled edge	A 0.8 mm bevel round the foot of the base, so a scraper gets under it and the model comes off the build plate without chipping.
Notches	Half-round notches at the foot of the wall, front and sides, so a lab can seat the model the same way each time.
Tie bar	A bar joining the two back ends of a full arch at the floor, so it cannot spread or warp while it prints. Cut it off afterwards [2].
Follow gum line (Aligner models only, on by default)	Keeps an even 2 mm band of gum below the teeth all round, about 18 mm tall overall, and removes the palate and floor of the mouth. Best for thermoforming.

Inside the model

Choice	What it makes
Hollow	A shell with an empty inside: the least resin.
Honeycomb	A shell filled with a honeycomb: stiffer, still light, and it drains.
Solid	Solid all through: the most accurate, and the heaviest.

For a hollow or honeycomb model, also choose:

- **Wall thickness**: 2, 3, 4 or 5 mm. The default is **3 mm**. Hollow models lose accuracy as the shell gets thinner, and 3 mm shells stay within clinically acceptable limits for the full arch on resin printers [3]. Use 3 mm or more for thermoforming [1].
- **Base underneath**: **Open** leaves the hollow open underneath to drain and clean. **Honeycomb** adds a honeycomb across the bottom to stiffen the base.



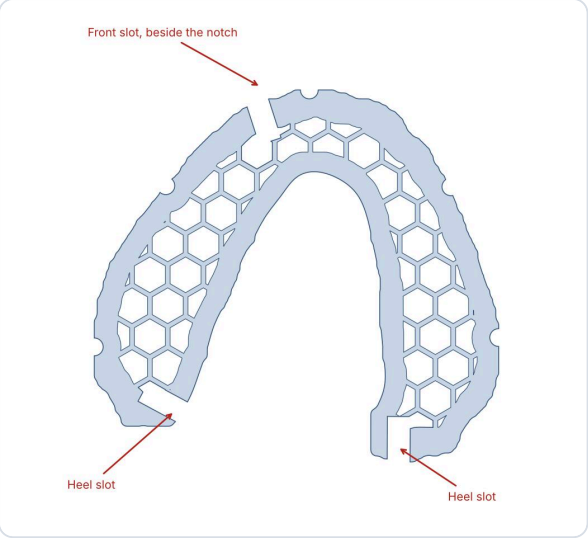
Aligner models: Follow gum line appears among the model options.

8 Hollow models and drain slots

Every hollow and honeycomb model is made with three drain slots at its base. Nothing needs to be set.

Why they are needed

Dental models print best flat on the build platform [4]. A hollow model printed that way is closed underneath by the platform, and once the top layers close over, its inside is a sealed cup. Each time the platform lifts to peel a layer, the resin and air inside are pulled and pushed. This is the "suction cup" or cupping effect. It can distort the model, pull the wall in, or blow it out [4-6]. Formlabs, Photocentric and Medit all add drain holes at the base of hollow models for this reason [4-9].



A Study model made from the scans in this guide, sliced 1.5 mm above the build platform. The honeycomb cells join one another, and the three slots join the inside to the outside.

What is made

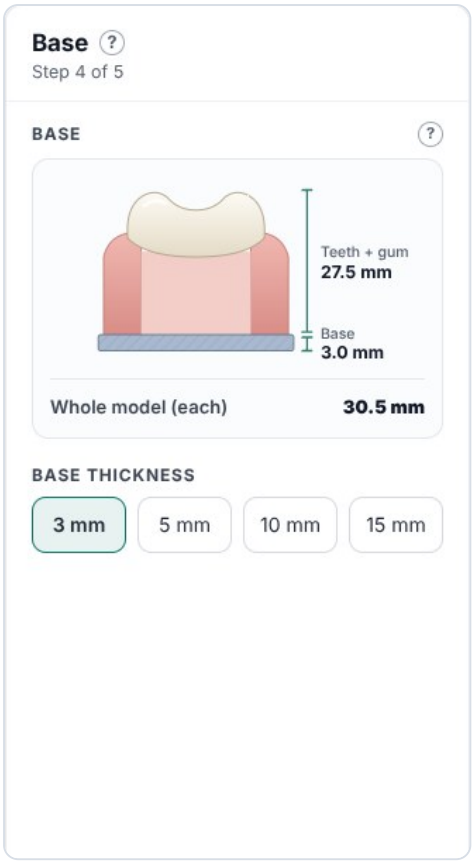
- **Three slots**: one at the front of the arch, one through the back of each heel [8].
- **Arch-shaped, 4 mm wide and 3 mm tall**, flat along the bottom so the resin drains out completely, and rounded on top. That is wider than the 3.5 mm minimum drain hole Formlabs specifies [5].
- **At the floor**: on the platform for a model open underneath, on the top of the floor for one with a closed floor [8, 9].
- **Through the outer wall only**, square to the wall, aimed at the hollow so the slot always reaches it, even at a tapered heel.
- **Clear of the front notch** and of a label, and never filled in by a notch, collar or tie bar.
- In a **honeycomb**, small slots at the foot of every cell wall join the cells, so all of them drain to the three slots.

Solid where accuracy matters most

Hollow models are not recommended where the highest accuracy is needed [4]. Crown-and-bridge and aligner models are solid by default.

9 Step 4 - Base

Set how thick the base is. The scan is used as you trimmed it.



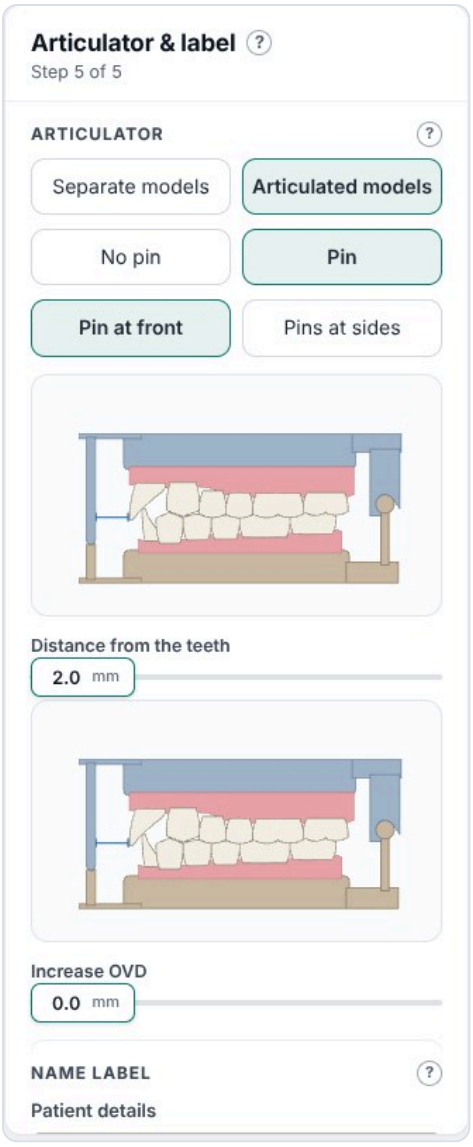
The Base panel: a cut through a molar, with the heights.

1. The picture shows a cut through a molar: the tooth and gum as scanned, and the base under them, drawn to the thickness you choose.
2. Choose the **Base thickness**: **3, 5, 10 or 15 mm**. The default is 3 mm.
3. Read the heights beside the picture: **Teeth + gum** (from the scan), **Base**, and **Whole model (each)**.
4. Press **Next: Articulator & label**.

A thicker base makes a stiffer, taller model and leaves room for a label on the side of an orthodontic base. A thinner base saves resin and printing time. For aligner models, the height is set for thermoforming at the Model step.

10 Step 5 - Articulator & label

Choose separate models or a printed articulator, name the models, and make them.



Articulated models with a pin at the front. The picture above each slider shows what it does.

Articulator

- **Separate models**: an upper and a lower, each on its own base.
- **Articulated models**: the pair with a printed articulator. A ball condyle behind each last molar drops into a cup under the upper from behind. The cup is a tenth of a millimetre bigger than the ball all round, so it turns smoothly with very little play. Nothing snaps, and no bands or clips are needed. To take the models apart, open them and lift the upper back off the balls.

Pin

- **No pin** (the default): the upper turns on the condyles as a hinge, and the teeth stop it closing, at the scanned bite.
- **Pin**: a split pin holds the bite. The upper half's rounded end rests in a shallow dish in the lower half, which centres it every time [17].
  - **Pin at front**: one pin in front of the incisors. The most accurate for the vertical dimension.
  - **Pins at sides**: a pin each side in the premolar region, so the front teeth are clear to see, photograph and seat an index on. Any height error in a printed pin shows about twice as large at the incisors.
- **Distance from the teeth**: how far the pin stands from the nearest tooth, 2-10 mm.
- **Increase OVD**: opens the bite at the pin, raising the occlusal vertical dimension by this much at the incisors, 0-10 mm in 0.5 mm steps [16].

Name label

Type the **Patient details** (a name or code, up to 24 characters). It is engraved on the base, and it starts the names of the saved files.

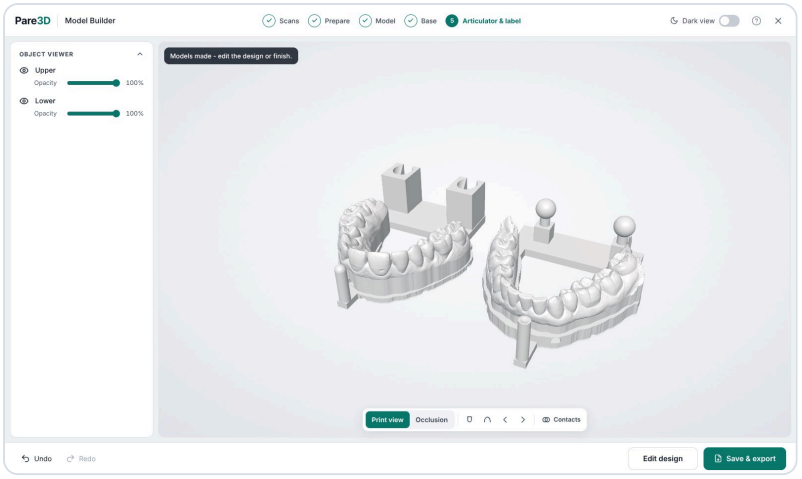
Press **Make models**.

A hinge, not a jaw

The printed articulator reproduces the closed bite and opens on a fixed hinge. It does not record or reproduce jaw movements, condylar guidance or a facebow transfer.

11 Making, checking and saving the models

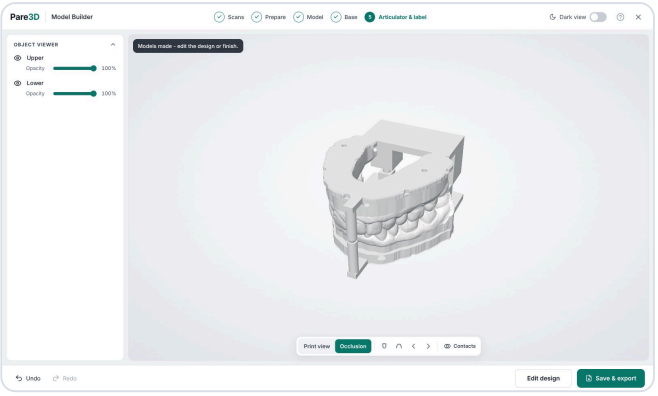
While the models are made, the Next button shows *Making the models…* and **Stop** cancels without changing anything.



The models made, in Print view: laid out as they will print.

Check them

- The panel closes so the models fill the screen.
- **Print view** shows them laid out to print. **Occlusion** shows them in the bite.
- **Contacts** works on the finished models, in either view.
- The Object viewer lists the made models, with an eye and opacity for each.



Occlusion: the pair in the bite.

Save and export

1. Press **Save & export** and choose where to save. The button fills green and reads *Saving…*.
2. Two STL files are saved, the upper and the lower, laid out as in Print view:
   - **RJ - Upper - 2 Nov 2026.stl** and **RJ - Lower - 2 Nov 2026.stl** with the label *RJ*.
   - **Upper - 2 Nov 2026.stl** and **Lower - 2 Nov 2026.stl** without a label.
3. The button changes to **Exit**. Press it to return to the Dental Studio page. The case card now reads *Models made*.

Not quite right? Edit design

Edit design brings back the scans and every setting as they were, so you can change anything and make the models again. You can also open the case from its card at any time.

12 Printing

Open the saved STL files in your printer's slicer (for example PreForm, CHITUBOX or Lychee).

Topic	Recommendation
Layer height	25-50 µm (0.025-0.05 mm). Coarser layers leave lines across the tooth surfaces that a silicone index copies.
Orientation	Flat on the build platform, base down, as laid out. It is the most stable and accurate way to print a model, needs no supports, and saves layers and resin [4].
Hollow and honeycomb	Print flat. The drain slots let resin in and out during printing and washing (chapter 8). Check in the slicer that the slots sit on, or just above, the platform. Wash long enough for the solvent to flush the inside, and let it drain out of the slots before curing.
Articulator	Print flat, as laid out. Keep supports off the contact surfaces: the cups under the upper, the top half of each ball, and the pin's rounded end and dish. Printed flat these face away from the platform and need none. A support mark there changes where the models close. Use your printer's finest validated layer height.
Removal	Keep **Bevelled edge** on. It gives a scraper an edge to get under.
Tie bar	Cut it off after curing.

Before printing a hollow model you have changed outside Pare 3D

If a model is re-hollowed or re-based in other software, check that it still has drain holes at its base. A hollow model without them is a sealed cup when printed flat [4-6].

13 Messages and what to do

Message	Meaning and what to do
One scan must be the upper and the other the lower	Both boxes name the same arch. Use the swap button or the lists at step 1.
The two scans are not where the scanner put them...	The arches do not share coordinates (one was moved or turned in other software). Each is made on its own, without bite or articulator. Re-import the original exports.
The arches do not touch: add the bite scan...	No bite scan, and the arches are apart. Add the bite scan at step 1. Without it the models are made separately.
A bite scan does not match these arches well...	A bite scan disagrees with the scanner's bite by more than 0.5 mm. Check the occlusion from the front and both sides, and with Contacts. Rescan the bite if it is wrong.
... of the selected triangles are on the crowns of the teeth	Trim stopped before cutting into the teeth. Unselect them, or press again to go ahead.
Part of the scan's ragged edge could not stand on the base...	The scan was cut where its edge is complete. Check that nothing you need was lost, and rescan deeper if it was.
Some of the tooth surface could not be kept where the scan's edge folds under the teeth	An undercut or a fold at the edge. Trim the fold at step 2 and make the models again.
There was no room for vent holes	A hollow model with a closed floor is too small for its floor vents. The drain slots are still made. Use **Open** underneath or **Solid** if you prefer.
The label was left off...	There was no room for letters at least 1.5 mm tall. Use a shorter label or a thicker base.
Teeth of one arch meet the other's gum in the bite (... mm²)	For example, lower incisors on the palate in a deep overbite. Left untouched, so the printed models show the contact. Note it for the treatment plan.
Soft tissue cut back clear of the other model: ... mm²	Retromolar or tuberosity tissue that would have met the other model was trimmed 0.5 mm, as a technician trims a cast's heels.
The articulator opens less than 30° before the models meet	Something blocks the opening (tall distal tissue, a tie bar). Check it from the side before printing.
Part of a model came out loose from its base	With an articulator: do not print, and report it with the scans (**Report a problem** at the foot of the Dental Studio page). Without one: a piece of scan stands free. Trim it at step 2 and make the model again.
Stopped making the models. Nothing was changed.	You pressed Stop. Change what you want and make them again.
The scans changed while the models were being made	A scan was edited during making. Make the models again.

14 Quick reference

<table>
<tr><td>The five steps</td><td>1 Scans · 2 Prepare · 3 Model · 4 Base · 5 Articulator & label. Then Print view / Occlusion, Save & export, Exit.</td></tr>
<tr><td>Defaults by model</td><td><table>
<tr><th>Model</th><th>Base</th><th>Inside</th></tr>
<tr><td>Study model</td><td>Clean, 3 mm</td><td>Hollow, honeycomb base, 3 mm walls, drain slots</td></tr>
<tr><td>Ortho study models</td><td>Orthodontic, trimmed</td><td>Solid</td></tr>
<tr><td>Aligner models</td><td>Follow gum line, about 18 mm tall</td><td>Solid</td></tr>
<tr><td>Crown & bridge</td><td>Plate, 5 mm</td><td>Solid</td></tr>
</table></td></tr>
<tr><td>Base thickness</td><td>3 (default), 5, 10 or 15 mm.</td></tr>
<tr><td>Drain slots</td><td>Three, 4 mm wide × 3 mm tall, arch-shaped, at the floor: front, and back of each heel. Hollow and honeycomb models only.</td></tr>
<tr><td>Articulator</td><td>Ball condyles in cups, a tenth of a millimetre clear all round. No pin (default), a pin at the front, or pins at the sides. Distance from the teeth 2-10 mm. Increase OVD 0-10 mm in 0.5 mm steps.</td></tr>
<tr><td>File names</td><td>"Label - Upper - Date.stl" and "Label - Lower - Date.stl", or without the label if none is given.</td></tr>
<tr><td>Mouse and keys</td><td>Right or middle drag: turn the view · Wheel: zoom · Right-click: views and tools for the step · <code>Ctrl</code> + <code>Z</code> : undo.</td></tr>
</table>

15 Accuracy, and checking a printed articulator

Accuracy

The Model Builder does not change the scanned tooth surfaces. Bases, pins and the articulator are added around the scan, and the teeth are carried through as the scanner recorded them. The accuracy of a finished model therefore depends mainly on:

- **The intraoral scan.** Trueness and precision are set by the scanner, the scanning technique and the clinical conditions. Refer to your scanner manufacturer's specifications.
- **The bite record.** The Model Builder keeps the jaw relationship exactly as your scanning software exported it. Digital bite records are generally more accurate than silicone or wax records [10]. Bilateral buccal records are the most accurate across the arch [13-15]. Accuracy falls with fewer teeth, quadrant records, large edentulous spans, and with distance from where the bite was scanned [11-13].
- **The model type.** Solid models are the most accurate. Hollow models lose accuracy as the shell gets thinner [2, 3], which is why hollow walls default to 3 mm.
- **The printer, resin and post-processing.** Dimensional accuracy depends on the printer's calibration, layer height, resin, orientation, washing and curing. Follow the resin manufacturer's validated settings.
- **The articulator.** A printed hinge reproduces the closed bite only (chapter 10). Hand articulation of printed casts has been reported to be more accurate than CAD-generated articulation pins [18]. Check the printed occlusion against the scans before relying on it.

Before you use a printed model for any restoration, appliance or diagnosis, check it. Compare the on-screen occlusion with the patient's scans, confirm that the printed teeth seat correctly against each other, and verify critical dimensions where they matter.

Checking a printed articulator

1. Close the models on the hinge. They should close without forcing, and the condyles should turn smoothly in their cups.
2. Check the occlusion against the on-screen Occlusion view with Contacts on: the same teeth should touch.
3. With a pin, check the pin's rounded end sits in its dish with the teeth in contact (or, with Increase OVD set, at the set opening at the incisors).
4. Open and close several times. The bite should return to the same place each time. The ball-in-cup seats locate the upper on the condyles the same way every time [17].
5. If the bite does not match, check the print for support marks or residue on the cups, balls, pin and dish, and reprint at a finer layer height if needed.

16 References

Model types and accuracy (1-3), printing hollow models flat and drain holes (4-9), the bite and virtual articulation (10-16), the articulator (17, 18) and records (19).

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Pare 3D is not a medical device. Check every model before clinical use.